But, because of the large population of the country, arable land per capita is not high; the figure of 0.14 hectares is lower than the average for the world (0.22) and is only one-fourth of the U.S. (0.59) figure. About 15 per cent of the sown area is multi-cropped (sown more than once in a year), while one-fourth of the gross cropped area is irrigated. Most of the multi-cropped area is irrigated and the security provided by irrigation facilities is a major factor in intensive application of labour and inputs to obtain high yields.

B) **Non-agricultural Land**

This includes land under forests, permanent pastures and other non-agricultural uses (towns, villages, roads, railways, etc.) and land classified as cultivable waste as well as barren and uncultivated land of mountain and desert areas.

Three important changes in the land utilisation witnessed during the last seven decades are:

- reclamation of waste and fallow lands
- a significant increase in the 'area sown more than once'.
- a significant fall in the land area under cultivation.

Perspective. It is clear that the total supply of land is a fixed factor. Therefore, what is required is that an effective rationing of land among the varied uses be made. As far as possible, no further encroachments on cultivable land should be allowed; priority should be given to the use of non-cultivable land for non-agriculture uses. This will not only save cultivable land for agriculture but will also promote a balanced regional development.

4.2.2 Issue of Land Acquisition

With increasing population, explosive urbanisation and growing industrialisation land resources are increasingly coming under stress. The issue merits serious attention and calls for satisfactory solution.

Compulsory land acquisition continues to foment profound distress and often violent resistance in many parts of the country. At the heart is a colonial law of 1894, which equipped government with powers to compulsorily acquire land at low costs. This was widely used to coercively obtain land from millions of usually impoverished, and often tribal, landholders, for large dams and public infrastructure projects. Their pauperisation led to democratic as well as militant struggles which rocked many parts of the country.

But public discontent grew further in the last three decades, when governments used this power to compulsorily acquire arable lands for private industry and real estate developers. A politically volatile – and ethically complex – debate has grown around whether governments should use the power vested in it for public good, to force landowners to sell their land to private industry.

There is a larger context to it, including concerns of food security. The policy needs to be based on a fair and just reading of the Doctrine of Eminent Domain. When the state recognises private property, the private owner is the absolute title holder. However, the state is the paramount title holder. Should it need a piece of land for a clear and apparent public purpose – building a highway, for instance – it can resort to land acquisition. Of course, this should be done with generous compensation and not just a textual interpretation of 'market rate'.

The Doctrine of Eminent Domain cannot be misused to acquire land for favoured industrialists or real estate companies. That would be a gross misuse. As we saw in West Bengal in the Left Front years, it would lead to crony capitalism and sweetheart deals. If private players want to build factories or condominiums, they must buy from the farmer directly.

However, it is not as if private companies should be free to buy agricultural land at will. As a pre-requisite to a new land acquisition law, India needs a proper, digital, and easily available (it must be online, for instance) map of its entire land area. This map must establish exactly which tracts of land are fertile and suitable for multi-crop cultivation. Such land is a national asset. It is critical to growing food for our 1.3 billion people and serve as a bulwark for food security. Therefore, it must be a 'no go' area for industry.

One view is that governments should not coerce landowners to part with their lands for profit-led corporate entities. Let private industry negotiate with landowners, and give them what they demand, if they voluntarily agree to sell their lands. The problem with this view is that it would amount to exposing powerless farmers to highly unequal negotiations with powerful corporate bodies, which might arm-twist or short-change them. In addition, a just land acquisition law needs to recompensate agricultural workers and tenants; provide land for land, jobs, and a house. None of these would be on offer by unregulated purchases by private industry.

A fair deal for all persons who will be displaced by industry can be ensured only if all or most affected persons first freely consent to the purchase. It should then be mandatory for industry to approach government, at least above a certain threshold and for public purpose; and for government to ensure that they get paid fairly that the landless and tenants are also compensated, and all are humanely rehabilitated.

The country is in the throes of potentially explosive competing land hunger of peasants and large industry. It is the duty of the state to ensure that displacement is minimised, and all displaced persons get a fair deal. It cannot do this by just standing by and hoping markets will protect the poor.

4.2.3 Need for a Comprehensive Land-use Policy

Although India is one of the fastest growing economies in the world, its growth potential has been compromised by resource misallocation, especially when it comes to land. India is one of the most land-scarce countries in the world, and demand for land has accelerated with the increase in the pace of industrialisation and urbanisation. But huge distortions in land markets have slowed the pace of growth. If these resources could be used more efficiently, India has the potential to achieve double-digit growth.

Conventional wisdom has focused on the labour market as being the most distorted in India. But there are even bigger distortions in the other factor markets.

Distortions in land markets are much bigger than those in labour markets. A comparison of factor misallocation indices at the district level has shown that an increase in the misallocation of all factors is associated with a huge decrease in output per worker in the manufacturing sector. Most of this decline originates from the misallocation of land and buildings. This appears to be at the root of much of the misallocation of output, and it accounts for a large share of the differences in productivity.

If land is so highly misallocated, this has repercussions on capital allocation through financial markets.

The two are interconnected. Most bank loans require some form of collateral to guarantee the loan. Land is simply the best form of collateral due to its immobility (i.e. the debtor cannot run off with land). While borrowers can often pledge 80 per cent of the land value against loans, for most other forms of fixed investment, the loan-to-collateral value ratio is substantially lower.

Misallocation in labour market inputs has no adverse impact on the allocative efficiency of financial loans. When it comes to land misallocation, however, the consequent degree of financial misallocation has only worsened over time as large manufacturing firms have moved out from cities and into rural areas in search of more land.

Distorted land markets are a breeding ground for crony capitalism and political subsidies. While the policy focus on improving land administration and regulation is well-placed, there are bigger growth benefits that can be derived from shifting the policy focus from reducing land "regulatory tax" to increasing land revenue tax. This will enable more efficient firms to grow faster and increase the budgetary revenue to maximise finance for development, and additional revenues needed for investments in infrastructure, urbanisation, housing, and social programmes.

Broadening the tax base will not only enable India to improve efficiency in resource use and accelerate growth, but it will also make growth more inclusive.

4.2.4 Soils

Long ago, Aristotle described soil as the stomach of the plant. Even now over 90 per cent of the world's food comes from the soil and less than 10 per cent comes from both inland water and the oceans.

The cropping pattern of the country is greatly influenced by the soils and the elements of the physical environment. The Indian Council of Agricultural Research divides the soils found in the country into eight major groups which are: (i) Alluvial soils including the coastal and deltaic alluvium; (ii) Black soils of varying types; (iii) Red soils, including red loams; (iv) Laterite and lateritic soils; (v) Forest soils; (vi) Arid and desert soils; (vii) Saline and alkaline soils; and (viii) Peaty and organic soils. Keeping in view their extent and agricultural importance, the first four, *viz.*, alluvial, black, red and laterite soils in that order, form the most important soil groups in the country. Almost the entire cultivated area in the country is covered by these soils.

- Alluvial soils are suitable for the cultivation of almost all kinds of cereals, pulses, oilseeds, cotton, sugarcane, and vegetables.

- Black soils are known for their fertility. They give good yields despite continued cultivation and without proper manuring. Cotton, cereals, oilseeds and many kinds of vegetables and citrus fruits are some of the crops suited to black soils.